

Missile launch Project Report

Ayush Baran Sen , Rupayan Chowdhury , Rahul Ganguly, Anjan Sadhukhan

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1. Abstract

In this work we study how much distance does a rubber band made of synthetic material travel when it is pulled to a certain extent. The problem statement is ***Missile launch planning***: This impressive project is somewhat marred by the fact that the missile involved is just a rubber band. The launcher consists of a ruler along which the band is pulled to some known extent before being released. Your job is to collect data and come up with a formula that tells you how much to stretch in order to reach a given distance. You must also give some error bound. After collecting the data we used ordinary least square linear regression to come up with the formula that links the travelled distance and pulled length. With this report we also attach a python program that will calculate necessary stretched length, so that the rubber band can travel a given distance.

2. Acknowledgements

We would like to extend our heartfelt appreciation to everyone who contributed to the successful completion of this project. Our sincere thanks go to our project team members for their dedication and collaboration throughout the project. Each member played a significant role in shaping the outcome. Special thanks to our professor, Dr. Arnab Chakraborty, for his guidance and valuable feedback, which enriched our work.

3. Necessary Equipments

We used a bunch of rubber bands, a wooden scale, a chair, a 15cm plastic scale, a measuring tape to conduct the experiment. Then the gathered data was analysed in R on a computer.

4.Introduction

Almost all of us have played with rubber bands as it can be used to hit something far away by stretching it to some length and then releasing it. In this project we try to create a mathematical model that relates amount of distance travelled by a rubber band by stretching it some amount. Our model will tell us how much to stretch the rubber band so that it travels a specified length. Initially, We did not have any idea how the model would look like. So as a first step we gathered the data . Then we plotted it . The plot looked like a linear equation. So, we decided to go with linear regression to come up with a mathematical equation. The next sections give details of the project's execution and necessary plots to support our conclusions we have made from the data.

5.Conducting the experiment

At first, We took just one rubber band and decided to use that as the representative rubber band of the population of rubber bands. We took several measurements using the rubber band. After gathering several data points we saw that the rubber band was no longer useful to gather data. This led us to conclude that only one rubber band can not be used to gather data. Because as time passes and we conduct experiments using a rubber band , the rubber band suffers wear and tear. This leads to error that can be controlled if we use several rubber bands to gather data. So, this time we decided to use 9 rubber bands selected randomly from the bunch of rubber bands. We decided to use each rubber band for a set number of times to gather data. At specific pulled length we took two measurements for each rubber band. Then we averaged the data to associate that average as travelled distance with that specific stretched length. We did this close to 180 times. We took data in a manner so that we can identify which data came from which rubber band. After completion of data gathering stage we did some computations on the data to calculate actual stretched length. We plotted the data using R. For most of the rubber bands the corresponding plot suggested linear relation. This suggested us to use linear regression to model it. As we wanted to use our model for any rubber band from the population we used, We mixed the data and plotted it. The data cloud still contained the essential feature of being linear like. So seeing this we went ahead and used linear regression.

6. Using R to analyse the data

We compiled the data so that we can work on it. At first we plotted the stretched length(x) vs distance travelled(y) for every Rubber band. We saw a linear pattern for each rubber band. Here are the graphs for all the Rubber bands:

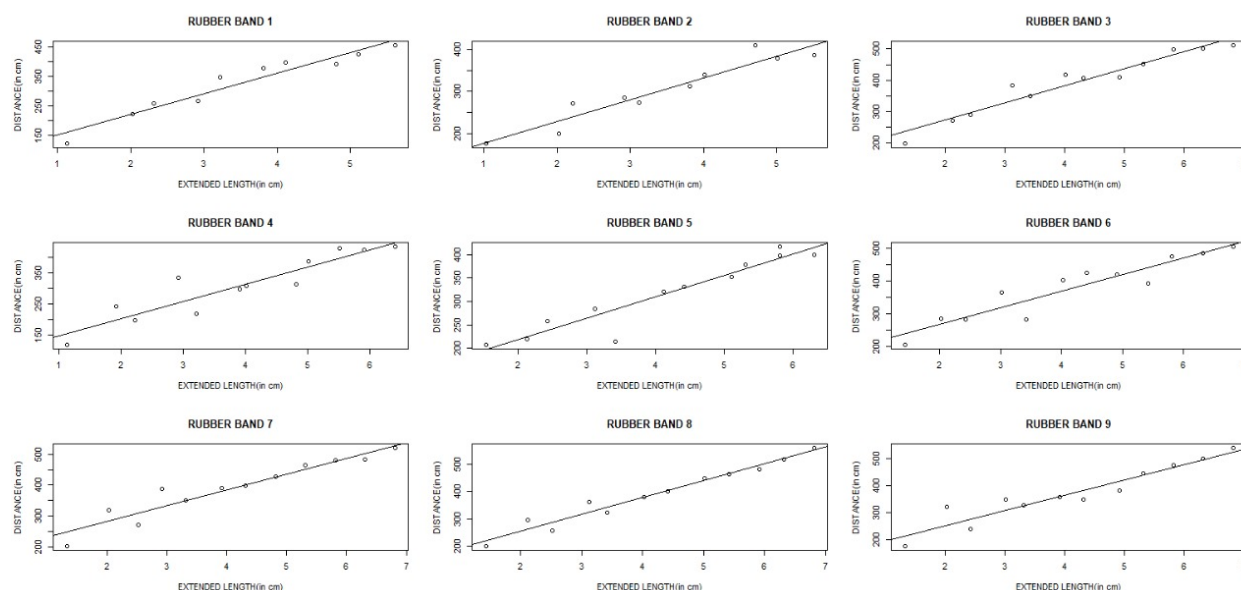


Figure 1: Graph of extended length vs distance for each rubber band

So, we have already seen the individual extended length-distance graphs for each rubber band. Now we are going to see the overall(pooled data graph) of length -distance.

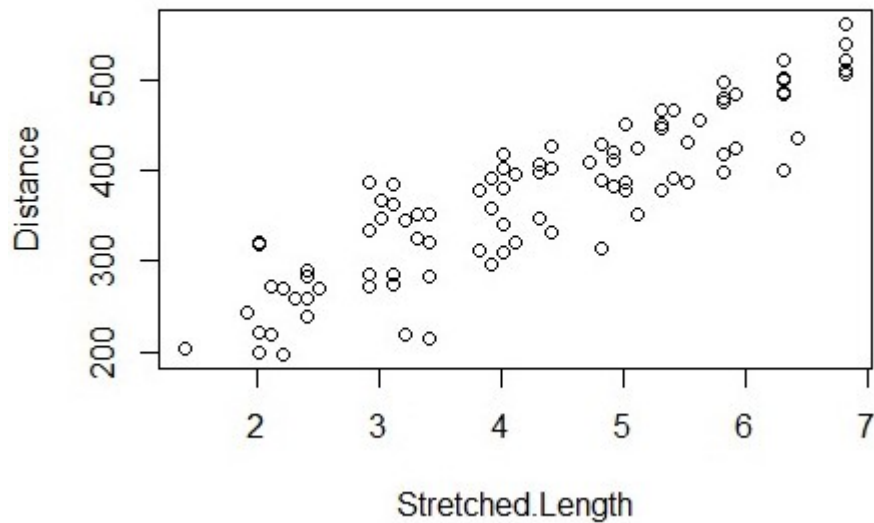


Figure 2: Pooled data graph

Here we can see that the scatter plot seems roughly linear in nature. So, We use `lm` command to fit a Ordinary least square model of the data. When we fit data using a modelling technique we assume things about the data and variables. For linear regression those assumptions are as follows:

- **Linear relationship:-** There exists a linear relationship between the independent variable, x , and the dependent variable, y i.e. y is some one degree polynomial of x .
- **Independence:-** The residuals are independent. In particular, there is no correlation between consecutive residuals in time series data.
- **Homoscedasticity:-** The residuals have constant variance at every level of x .
- **Normality:-** The residuals of the model are normally distributed. If one or more of these assumptions are violated, then the results of our linear regression may be unreliable or even misleading.

Assumption 1: Linear Relationship:-

Explanation:-

So the first assumption of linear regression is to show linearity between the the extended length and the distance which for plots of 9 rubber bands. We need to show that in which plots we can fit a straight line. In this plots figure 1 rubber bands 1,3,5,7,8,9 satisfy a good linear relationship between the variables and figure 1 rubber bands 2,4,6 do not satisfy good linear relationship between the variables as other ones. Although our goal is to see if the pulled data satisfy good linear relationship between the variables as other ones. We have already seen that the data cloud has roughly linear shape. so the first assumption is met.

Assumption 2: Independence:-

Explanation:-

The simplest way to test if this assumption is met is to look at a residual time series plot, which is a plot of residuals vs. time. Ideally, most of the residual autocorrelations should fall within the 95% confidence bands around zero, which are located at about $\pm 2/\sqrt{n}$, where n is the sample size. Though we will not try to observe this assumption since our data is not a time series data.

Assumption 3:Homoscedasticity:-

Explanation:-

To prove that the data is Homoscedastic we first have to show that it is not Heteroscedastic. These can be shown in several ways as follows:

- **Transform the dependent variable:**

One common transformation is to simply take the log of the dependent variable. For example, if we are using population size (independent variable) to predict the number of flower shops in a city (dependent variable), we may instead try to use population size to predict the log of the number of candy stores in a city. Using the log of the dependent variable, rather than the original dependent variable, often causes heteroskedasticity to go away.

- **Redefinition of the dependent variable:**

One way to redefine the dependent variable is by defining it with respect to some variable i.e. rate. As for example, instead of using the population size to predict the number of candy stores in a large city, we may instead use population size to predict the number of candy stores per sq km in a large city. In most cases, this reduces the variability that naturally occurs among larger populations since we're measuring the number of candy stores per neighbourhood, rather than the sheer amount of candy stores.

- **Use weighted regression :**

Another way to fix heteroscedasticity is to use weighted regression. This type of regression assigns a weight to each data point based on the variance of its fitted value. Essentially, this gives small weights to data points that have higher variances, which shrinks their squared residuals. When the proper weights are used, this can eliminate the problem of heteroscedasticity.

- **scale-location plot:** There is another way to see homoscedasticity is to look at Scale-Location plot of the model. This is an informal method. If the curve drawn is roughly linear, then we can say that the homoscedasticity assumption is likely satisfied.

Assumption 4: Normality:-

Explanation:-

The next assumption of linear regression is that the residuals are normally distributed. We will check it by drawing a histogram and see if it fits normal distribution. It can be checked by QQ residual plot if the points on the plot roughly form a straight diagonal line, then the normality assumption is met.

So, after fitting the model we use summary command and plot command to get the summary and diagnostic plots of the model. We look at the scale-location plot of the model. Here is the scale-location plot of the model:

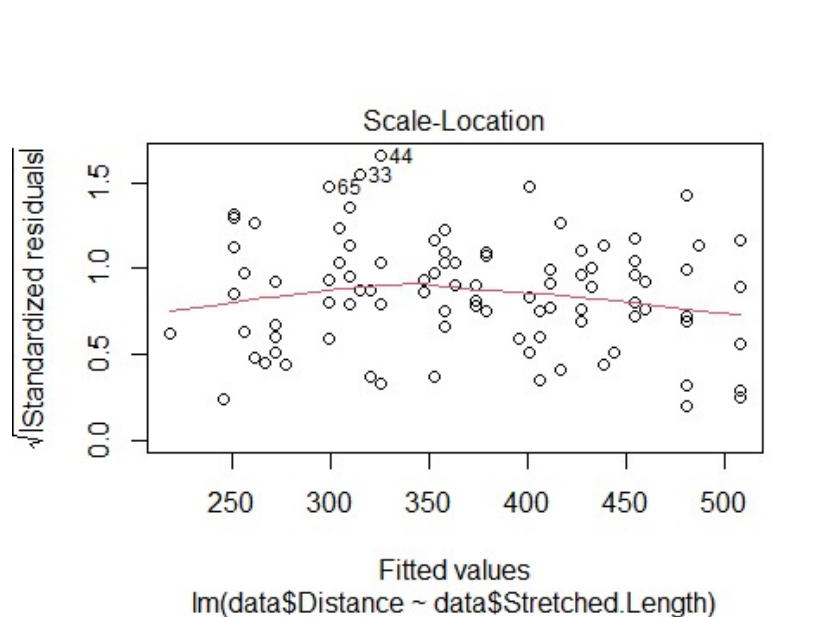


Figure 3: Scale - Location plot

This plot is used to check the assumption of equal variance (also called “homoscedasticity”) among the residuals in our regression model. If the red line is roughly horizontal across the plot, then the assumption of equal variance is likely met. we see that the red line here is roughly horizontal. So it seems that the homoscedasticity assumption is met. Now, we look at the residual vs fitted plot. Here is the plot:

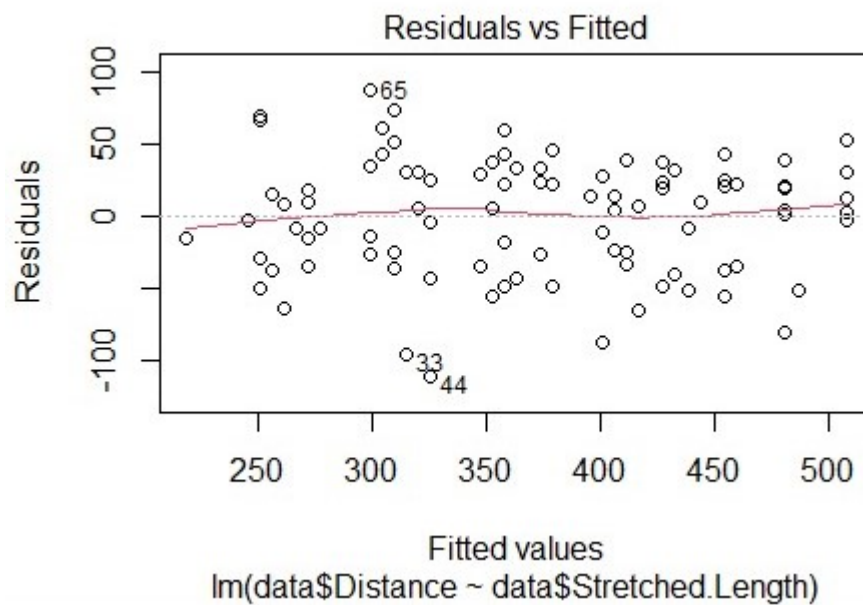


Figure 4: Residuals vs fitted plot

We see that the red line is roughly linear. Both of these observations show that a linear model can be used to fit the data. Now, lastly we check the Q-Q Residual plot. Here is the plot:

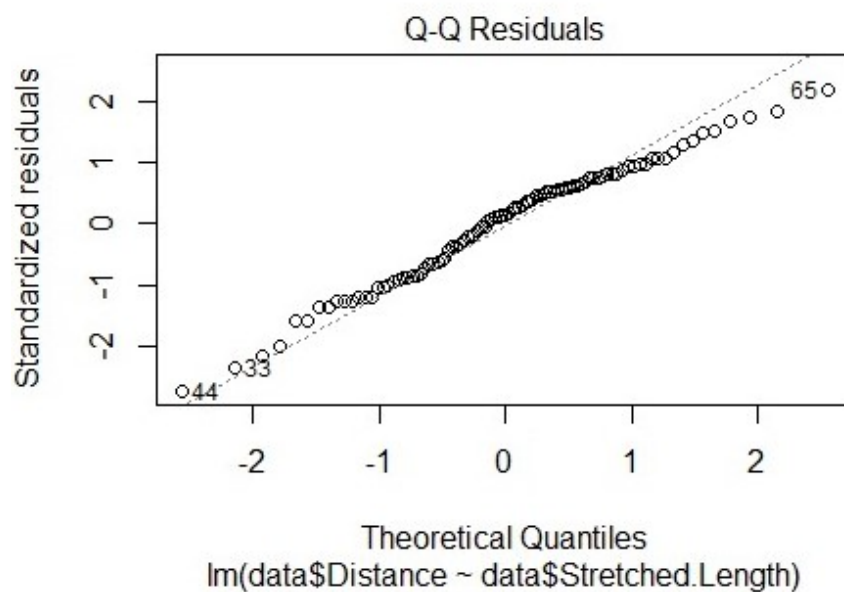


Figure 5: Q-Q residuals plot

We see that the graph is roughly linear. It has two wobbles at the start and end of the graph. Now we draw the histogram of the distance travelled by the rubber band. Here is the Histogram picture:

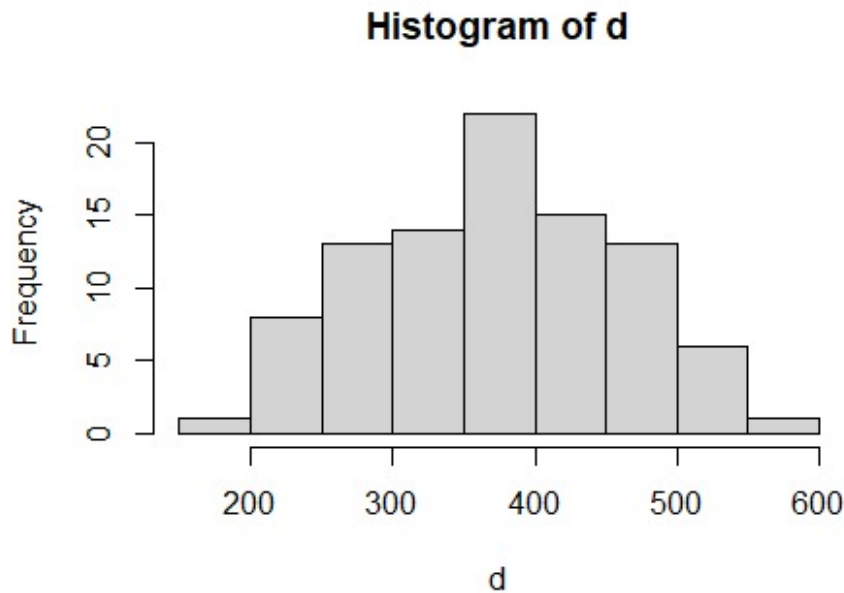


Figure 6: Histogram

As we can see that the important assumptions are met by model and data. So we may take this model as descriptor of the experiment results. Here is the summary of the model:

```
Residuals:
    Min       1Q   Median       3Q      Max
-110.860  -33.295   5.493   29.792   87.831

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    142.880    13.174   10.85  <2e-16 ***
data$stretched.Length  53.533     2.913   18.38  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.65 on 91 degrees of freedom
Multiple R-squared:  0.7877,    Adjusted R-squared:  0.7854
F-statistic: 337.7 on 1 and 91 DF,  p-value: < 2.2e-16
```

Figure 7: Summary of the linear model

Note that the calculated pearson correlation coefficient of extended length and distance is 0.8875362.

We see that R^2 is 0.78. So, this tells us 78 percent of the of the variation in the traveled distance can be explained by the stretched length. The RMSE of the model is 40.64939. The prediction interval and confidence interval are shown in the picture.below.

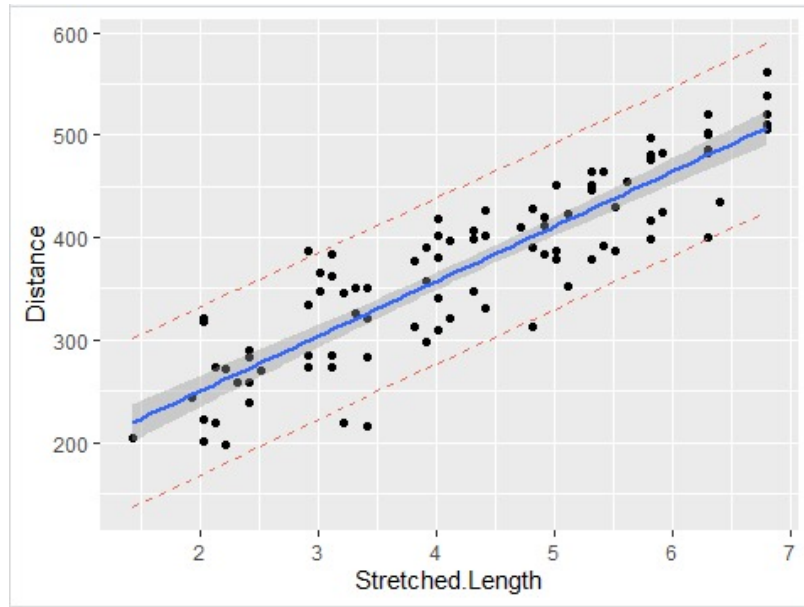


Figure 8: Prediction and confidence interval graph

7.Drawbacks of this experiment

There are several variables on which the experiment depends.But we do not have enough equipment to control every such variables.So we only took those variables in consideration which we can control.The drawbacks are as follows:

- Temperature plays an important role in varying the elasticity of the rubber band resulting different distance in same extension of the rubber band.This can be controlled by performing the experiment in a temperature constant room.We were not able to do that due to limited resources.
- Air resistance also happens to affect this experiment.It an be controlled by performing this in a vacuum room.But we were not able to do that due to limited resources.

- We used a chair to support the scale used to launch the rubber bands. And we ourselves release the rubber band by releasing hand over scale .This may not give full accuracy since our extension strength of band may change. Though it can be corrected by using a mechanical object like a robot.